

Jesús González-Ferrer

Computational Biologist & ML Scientist — foundation models, multimodal neural data, production ML pipelines
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SUMMARY

PhD graduate (Bioinformatics & Bioengineering, UC Santa Cruz) building deep-learning and foundation-model methods for single-cell genomics and multimodal neural data. First author of **SIMS** (Cell Genomics) and **HIPPIE**; led single-cell analysis for the SSPSYgene consortium at ~6M-cell scale and AI target-discovery research at Genentech. Core strengths: explainable single-cell ML, translational bioinformatics, multimodal and spatial data integration, cross-functional collaboration, scalability, and production ML pipelines.

PROFESSIONAL EXPERIENCE

Graduate Student Researcher — University of California, Santa Cruz (Braingeneers) 2022 – Present

PIs: David Haussler, Mohammed A. Mostajo-Radji

- Built **SIMS**, an explainable deep-learning model for cell-type classification from single-cell RNA-seq, published in *Cell Genomics* and released as open-source tooling adopted by other labs.
- Developed **HIPPIE**, a multimodal deep-learning model that classifies neurons from extracellular electrophysiology, published in *Nature Communications*.
- Designed and validated cross-modal frameworks integrating spatial transcriptomics with electrophysiology across organoid datasets.
- Built web applications (**CytoVerse**, **UCSC Brain Explorer**) featuring a scalable architecture that democratizes single-cell RNA and electrophysiology analysis, with deep-learning models at the core.
- Provided cross-functional leadership for the joint analysis group of the **SSPSYgene** consortium, supporting a large-scale perturbation data resource encompassing **~6 million cells**. Orchestrated a cross-laboratory integration and perturbation-analysis architecture across multiple scRNA chemistries (10x v2/v3, PIPseq) and diverse **perturbation** modalities (**CRISPR-Cas9**, **CRISPRi**, **Perturb-seq**), facilitating convergence with bulk RNA-seq, organoid and primary tissue data.
- Contributed to grant proposals to NIH, NSF, CIRM, UCOP, and BBRF, supporting \$50M+ in awarded funding and \$60M+ under review.

AI Researcher — Genentech (gRED), South San Francisco 2025 – present

Managers: Alsu Missarova, Héctor Corrada Bravo, Jason Vander Heiden

- Developed graph-based and machine-learning approaches built on single-cell foundation models to power drug discovery pipelines and downstream statistical frameworks for target identification, translational impact, and discovery of shared disease mechanisms across conditions.

Machine Learning Engineer / Analyst — SDG Group Iberia (Data Tech) 2021 – 2022

- Designed and built an ML framework in R used by 50+ data scientists.
- Automated data processes in Python with Apache Airflow; migrated SAS ETL projects to SQL, saving ~€1,500 per employee.

Earlier research

Machine Learning Researcher — Universidad Complutense de Madrid (2021): ML to study multiple sclerosis and its relationship with cognitive impairment (PI: José Luis Rodrigo Ayala).

Computer Vision Intern — VINDOO (2021): trained YOLOv4 vision models to help farmers optimize fruit-harvest timing.

Neuroscience Researcher — Instituto de Neurociencias, Alicante (2019–2020): studied ROBO and STIL in neurogenesis and cortical expansion (PI: Víctor Borrell).

SELECTED PUBLICATIONS

† First author.

† **González-Ferrer, J.**, Lehrer, J., O'Farrell, A., Paten, B., Teodorescu, M., Haussler, D., ... & Mostajo-Radji, M. A. (2024). SIMS: A deep-learning label transfer tool for single-cell RNA sequencing analysis. *Cell Genomics*, 4(6).

† **González-Ferrer, J.**, Lehrer, J., Alvarez-Esteban, B., Moreno-Ochando, A., ..., Haussler, D., & Mostajo-Radji, M. A. (2026). HIPPIE: A generative model for electrophysiological analysis across species, technologies, and modalities. *Nature Communications*. <https://doi.org/10.1038/s41467-026-76939-w>

† Currie, R., † **González-Ferrer, J.**, Mostajo-Radji, M. A., & Haussler, D. (2026). CytoVerse: Single-Cell AI Foundation Models in the Browser. *bioRxiv*. Preprint.

- † **González-Ferrer, J.**, & Mostajo-Radji, M. A. (2025). Toward automated and explainable high-throughput perturbation analysis in single cells. *Patterns*, 6(4).
- van der Molen, T., Spaeth, A., Chini, M., Hernandez, S., Kaurala, G. A., Schweiger, H. E., **González-Ferrer, J.**, ... & Sharf, T. (2025). Preconfigured neuronal firing sequences in human brain organoids. *Nature Neuroscience*.
- Mostajo-Radji, M. A., Leon, W. R. M., Breevoort, A., **González-Ferrer, J.**, Schweiger, H. E., Lehrer, J., ... & Pollen, A. A. (2025). Fate plasticity of interneuron specification. *iScience*.
- Geng, J., **González-Ferrer, J.**, Voitiuk, K., Seiler, S. T., Schweiger, H. E., Hernandez, S., ... & Teodorescu, M. (2025). Weighted graph for longitudinal tracking of neurons in cortical organoids on a high-density microelectrode array. *IEEE EMBC 2025*.
- Zabetian, Z., **González-Ferrer, J.**, Lehrer, J., Teodorescu, M., Haussler, D., & Mostajo-Radji, M. A. (2024). Single-cell RNA sequencing analysis using deep learning. *STAR Protocols*.
- Sano, T., Sampad, M. J. N., **González-Ferrer, J.**, et al. (2024). Internet-enabled lab-on-a-chip technology for education. *Scientific Reports*, 14, 14364.
- Vera-Choqueccota, S., Ehrlich, D., Luna-Gomez, V., Hernandez, S., **González-Ferrer, J.**, ... & Mostajo-Radji, M. A. (2026). Cloud-connected pluripotent stem cell platform enhances scientific identity in underrepresented students. *Stem Cell Reports*.

TECHNICAL SKILLS

Single-cell & genomics: Scanpy, Seurat, Monocle, scRNA-seq/scATAC analysis, DESeq2, GWAS, Bioconductor

ML & deep learning: PyTorch, TensorFlow, Keras, scikit-learn; variational autoencoders, CNNs, foundation models, representation learning, gradient boosting; computer vision, time-series

Multimodal & neural data: Electrophysiology analysis (MaxWell HD-MEA, Neuropixels), spatial-transcriptomics integration, Braingeneerspy, pyNWB, AllenSDK

Production ML / MLOps & cloud: Docker, Kubernetes, Apache Airflow, PySpark, WDL/Dockstore, SLURM, Terra, Google Cloud, OVH; Weights & Biases, Hugging Face, ONNX

Software & web: Python (adv.), R (adv.), SQL, JavaScript, Bash; Git/GitHub/GitLab; Django, Flask, React; viz (Matplotlib, Seaborn, Plotly, D3, Bokeh, ggplot2)

Wet lab: Tissue culture, immunohistochemistry, PCR/qPCR, cDNA library prep, aseptic technique

AWARDS & FELLOWSHIPS

Dissertation-Year Fellowship — UC Santa Cruz (2026–27)

Honorable Mention — UCLA Emerging Genomic Scientist Symposium (2026)

QB3 Fellowship — UC Santa Cruz (2025)

SSPSYgene Travel Award (2025)

EDUCATION

Ph.D., Bioinformatics & Bioengineering — University of California, Santa Cruz (Braingeneers) | 2022 – Present

M.Sc., Computational Biology — Universidad Politécnica de Madrid | 2020 – 2021

B.Sc., Biotechnology (High Academic Performance group) — Universidad Miguel Hernández de Elche | 2016 – 2020

SELECTED PRESENTATIONS

CytoVerse: A Browser-Native Platform for Scalable, Privacy-Preserving Single-Cell Analysis — Poster, scverse Conference, Stanford (2025).

HIPPIE: Multimodal Electrophysiological Classification of Neurons — Poster, SSPSYgene Consortium Meeting, UCLA (2024).

Data Standards for Single-Cell RNA — Talk, SSPSYgene Consortium Meeting, UCLA (2024).

Single-Cell RNA Automated Pipeline in Terra — Talk, SSPSYgene Consortium Meeting, NIH (2023).

Unraveling Cell-Line-Specific Differences with Deep Learning — Flash Talk, CEGS Meeting, NYU (2023).

A Minimal-Code Tool for Cell-Type Classification from Single-Cell Data — Poster, Society for Neuroscience (2022).

LANGUAGES

Spanish (native) · English (C1; TOEFL iBT 110) · German (basic)